

Original Report

Ten-year trends in safe radiation therapy delivery and results of a radiation therapy quality assurance intervention



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Abstract

Purpose: This study reviews our institutional error data and assesses the effectiveness of a policy implemented January 1, 2011, as a "no rushed treatment" initiative to avoid universal, large-scale replanning for all patients in the event that a treatment unit is down for ≤ 1 day.

Methods and materials: Radiation error data between January 1, 2004, and December 31, 2014, were reviewed to determine absolute delivery error rates. Variables were compared (using a χ^2 or Fisher exact test) before and after the policy change, including planning versus delivery error status and differences in error type. We also evaluated time of day in relation to therapist shift change, deviation from scheduled time, and weekend treatment as predictors of error using a test of proportions or χ^2 test.

Results: Treatment delivery error rate over the entire period was 0.18% per fraction; the rate before intervention was 0.24% and after was 0.08%, $P < .001$. For the 5 years for which detailed records were available (2010-2014), 109 delivery errors were reported. Delivery error rate was 0.09%; before intervention 0.15% versus after, 0.08% ($P = .005$) and 94% were level 1 errors. Fifty-six percent were primary planning errors and 44% were primary delivery errors. Before intervention, large-scale replanning occurred 18 times/year versus 4.5/year after, with 21% versus 12% of errors directly attributable to large-scale replanning. Fourteen error reports specifically implicated a rushed environment as causal. There was no significant difference in error rate based on time of day ($P = .631$). Error rates were higher for weekend simulation and treatments, 1.3% versus 0.09% per fraction ($P < .001$).

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Conclusions: Delivery error rates at our institution were similar compared with published series from other academic institutions. A significant improvement in delivery error rate was appreciated after implementation of a "no rushed treatment" initiative. A significantly higher error rate for weekend treatments was noted, warranting consideration of additional quality assurance measures. © 2015 Published by Elsevier Inc. on behalf of American Society for Radiation Oncology.

Introduction

Patient safety is of paramount importance in the delivery of radiation therapy. Despite advances in both treatment planning and delivery, preventable errors still occur. In the words of Dr William Mayo, "Deliver me from the man who never makes a mistake, and also from the man who makes the same mistake twice."¹ In an ongoing effort to educate and facilitate safer and higher quality radiation oncology care, the American Society for Radiation Oncology and American Association of Physicists in Medicine have developed the Radiation Oncology Incident Learning System, a secure online reporting system that will be used to collect radiation therapy near-misses and safety incidents. The intent of this initiative is to facilitate the reporting of errors in a nonpunitive environment and allow for a national analysis of potential sources of avoidable errors in the delivery of radiation therapy.²

Previous publications report rates of treatment-related incidents on the order of 0.06% to 0.70% per fraction.³⁻⁶ Errors in radiation therapy can occur at multiple levels of the treatment planning or delivery process including errors in prescription, dose calculations, imaging, patient setup, equipment such as accessory treatment devices or machine or software error.^{6,7} Finally, delivery of safe and effective radiation therapy requires the collaboration of multiple members of a team, including physicians, nurses, physicists, dosimetrists, therapists, and hospital administrators, who may play an important role in setting the tone for reporting and fostering a safety culture.

It has been shown that rushed treatment delivery and decreased planning time may be associated with increased error rates.⁷ An occasional, but inevitable, dilemma in the delivery of radiation therapy occurs when there is a treatment machine failure. This can result in extended machine downtime and may necessitate replanning for an alternate treatment machine. Previously at our institution, large-scale replanning (LSR) was routinely performed in response to treatment unit downtime, regardless of medical necessity. An increased incidence of errors was anecdotally noted on days on which LSR was performed. To improve safety and enhance workflow, a department-wide quality assurance policy was implemented on January 1, 2011, as a "no rushed treatment" initiative. The purpose of this policy was to avoid universal, LSR for all patients in the event that a treatment unit is down for ≤ 1 day when patient care would not be compromised:

- When it is determined that a treatment unit is down for a single day or less, the department policy is as such that

all patients scheduled on this unit will be cancelled until the treatment unit is repaired, unless it is determined to require a recalculation by the patient's attending or floor physician.

The aim of this study is to evaluate and report our institutional experience with radiation therapy delivery errors and to determine factors that may predict for increased incident rates. Second, we assess the differences in error patterns before and after implementation of the aforementioned policy.

Materials and methods

Radiation administration and incident data between January 1, 2004, and December 31, 2014, were reviewed to determine absolute delivery error rates. For patients treated after January 1, 2010, which corresponded with full implementation of Mosaic (Elekta, Stockholm, Sweden), an electronic medical record system (EMR), detailed treatment delivery data were available in the EMR. Although before this date, the error reporting process was the same, treatment delivery data including treatment times, patient setup information, and therapist notes were documented in paper charts now stored at an offsite facility. Differences in error rates before and after policy implementation were compared using a χ^2 test. This same test was also performed for the most recent 5 years only (through 2014) in an attempt to separate the potential effect a staged EMR implementation (April 1, 2007-December 31, 2009) may have had on error rates. Delivery errors were defined as any deviation in dose or dose distribution as delivered to the patient compared with intended dose or dose distribution. For each treatment delivery error, source of error was characterized as a primary treatment planning error if there was an error in prescription, error in documentation, error in patient setup description, or dosimetric error. Source of error was characterized as a primary treatment delivery error if, in the absence of a planning error, there was incorrect use of an accessory treatment device, incorrect beam angle, port, or patient setup. Level of deviation was recorded using the Radiation Event Scoring System⁸ and per state regulations.⁹ Information regarding treatment site, technique (intensity modulated radiation therapy versus 3-dimensional), pediatric versus adult patient, means of error detection and reporting, error date, and error types were collected. Variables were compared using a χ^2 or Fisher

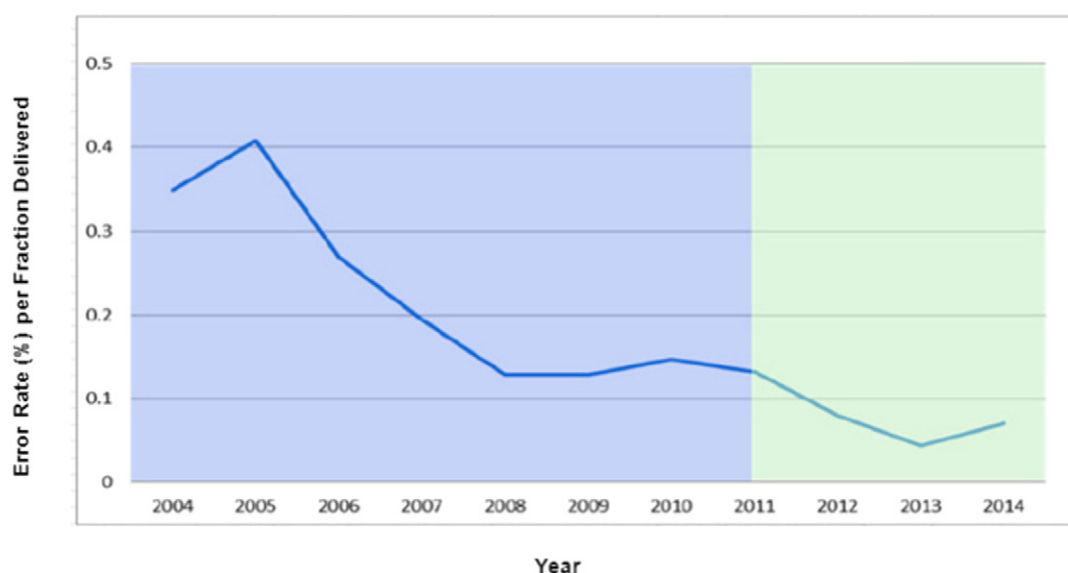


Figure 1 Institutional radiation error rate per treatment between 2004 and 2014. The blue background corresponds to the years before policy implementation; the green background corresponds to the years after policy implementation.

exact test before and after the policy change. A χ^2 test was used to evaluate time of day in relation to therapist shift change as a predictor of error; actual error rates were compared with expected error rates based on estimated machine-on time for morning (7:00-11:00 AM), afternoon (1:30-6:00 PM), and a break period (11:00 AM-1:30 PM) when therapists cycle off and back on. A test of proportions was used to compare timeliness of treatments for 100 randomly selected patients over the same period (2010-2014) versus timeliness of treatments when an error occurred, specifically the number of minutes between a patient's scheduled treatment time and documented treatment completion time. A χ^2 test was used to compare error rates for weekend treatments as compared with weekday treatments. All statistics were performed using R version 3.0.1 (The R Foundation for Statistical Computing, Vienna, Austria) statistical software.

Results

Over the period spanning January 1, 2004 through December 31, 2014, 250,568 treatments were administered and 461 treatment delivery errors occurred, resulting in an overall 0.18% error rate per fraction. Rate before implementation of policy was 0.24% and after implementation of policy was 0.08% ($P < .001$; Figure 1 and Table 1). For the 5 years (2010-2014) for which detailed records were available, 109 delivery errors were reported. Reporting was focused on errors reaching delivery, whereas near-miss reporting was optional and therefore not included in this analysis. For this period, overall delivery error rate was 0.09%; before intervention 0.15% versus after, 0.08% ($P = .005$). Ninety-four percent were documented Radiation Event Scoring System level I errors, defined as a solitary event that causes no harm to the patient and does not require a change to the radiation prescription. There were 6 level II deviations defined as an event requiring a change in the radiation prescription but not felt to pose harm to the patient. Two level II deviations were the result of incorrect shift from isocenter, 1 incorrect use of bolus, 1 incorrect field, 1 extra fraction administered, and 1 final fraction not administered. Fifty-six percent of errors were primary planning errors and 44% were primary delivery errors. Large-scale replanning occurred 18 times during the year before the intervention and dropped to 4.5 times per year averaged over several years following the intervention, with 21% versus 12% of errors directly attributable to large-scale replanning pre- and postintervention. Fourteen of the 109 error reports specifically implicated a rushed environment as causal. The most common error types were dosimetric and patient setup documentation, accounting for 35% of all errors. The

Table 1 Treatment deviations per fraction delivered

Year	Fractions delivered	Treatment deviations	Deviations per fraction delivered
2004	25,510	89	0.00348
2005	26,220	107	0.00408
2006	22,682	61	0.00268
2007	22,641	44	0.00194
2008	20,322	26	0.00127
2009	20,327	26	0.00127
2010	22,370	34	0.00151
2011	24,163	32	0.00132
2012	23,698	19	0.00080
2013	22,663	10	0.00044
2014	19,972	14	0.00070

Table 2 Total deviations per course of radiation therapy, 2010-2014

Treatment site	Deviations	Patients treated	Deviations per treatment course
Skin	5	74	0.067
Bone	13	760	0.017
Brain	7	635	0.011
H&N	5	724	0.007
Breast	45	909	0.050
Lung	11	902	0.012
Abdomen	2	249	0.008
Pelvis (includes brachytherapy)	17	1145	0.015
TBI	2	270	0.007
CSI	1	36	0.028
Extremity	1	82	0.012
Treatment type			
IMRT	11	1838	0.006
Non-IMRT EBRT	97	3737	0.026
Brachytherapy	1	211	0.005
Patient type			
Adult	104	5651	0.018
Pediatric	5	135	0.037
Total	109	5786	0.019

CSI, craniospinal irradiation; EBRT, external beam radiation therapy; H&N, head and neck; IMRT, intensity modulated radiation therapy; TBI, total body irradiation.

treatment sites where errors were most likely to occur were skin and breast, with 0.67 and 0.50 errors per treatment course, respectively. Table 2 and Figure 2 contain additional details regarding errors by treatment site, intensity modulated radiation therapy versus 3-dimensional, pediatric versus adult patient, means of error detection, and error type.

Review of a control population (100 patients) treated in our department over the same period as the study population revealed that 33% of treatments are completed with at least a half-hour deviation from scheduled start time. In our series, 43% of patients with a primary delivery error were completed, with at least a half-hour deviation from their scheduled time, which was not significantly different from the control ($P = .13$). Median deviation for control patients was 17 minutes versus 23.5 minutes for treatments with delivery error. For primary treatment delivery errors, there was no significant difference in error rate based on time of day ($P = .631$) (Figure 3). Error rates were significantly higher for weekend simulation and treatments, 1.3% versus 0.09% per fraction, ($P < .001$). Three were the result of dosimetric errors, 2 errors related to patient setup, and 1 error in transfer of plan data.

Discussion

In this study, we have analyzed the factors associated with errors in radiation delivery and examine the effect of a

department-wide quality assurance policy as it relates to our institutional error rate. When analyzing our outcomes for 2010 to 2014, we found that our overall rate for all reportable errors was 0.09% per fraction, which compares favorably to reports from other academic institutions.^{3-7,10} Of particular interest, however, is the observed decrease in error rate following implementation of a “no rushed treatment” policy in the event of treatment machine downtime. Directly following the implementation of this policy, our incident rate was nearly halved, from 0.15% to 0.08%. In the year preceding the policy change, there were 18 days during which LSR occurred, compared with an average of 4.5 days per year after the change.

LSR can stress departmental resources, particularly dosimetrists and therapists. Typically, machine downtime is unanticipated and staff is unlikely to be prepared to triage the replanning of large numbers of patient plans for an alternate machine in addition to routine planning of existing and new patients. When filing an error report at our institution, reporting individuals may provide an explanation and/or suggestion for preventing the incident in the future. To illustrate the potential stress induced by LSR, we provide examples as documented in the error reports: “Try not to be affected by patient wait time. Take necessary amount of time to replan,” “Take more time on patient recalculations regardless of patient wait time,” “...replanning many cases in limited amount of time,” “...machine down with multiple patients waiting,” “...dosimetry trying to minimize patient wait time while recalculating multiple cases,” and “...replanned many cases that day in addition to normal workload.” As evidenced by these excerpts taken from error reports submitted during LSR days, there is an inordinate amount of pressure applied to staff members as they attempt to replan and safely deliver treatment that directly correlated with higher error rates. Although treatment interruptions are not ideal, the potential for misadministration may outweigh the benefit of avoiding a missed fraction. Typically, when a treatment unit is down for an extended period, the missed fractions are supplemented by weekend treatments in an attempt to avoid lengthening the course of treatment, which has been shown to result in measurably decreased efficacy.¹¹ Additionally, patients for which the physician deems treatment interruption unacceptable can and will still be replanned.

Following the implementation of the no-LSR policy, we observed a decrease in departmental error rates, although this was not likely to be the only reason for the improvement over this period. Rather, the reduction in error rates observed over the period evaluated in this study is likely multifactorial in nature and improved automation and technological advancements are likely to play a role.¹² A potential weakness of our study is an inability to account for all technological advancements during this period. However, we did attempt to control for the independent effect of implementation of an EMR by separately analyzing both the improvement in error rate for the entire period and the past 5

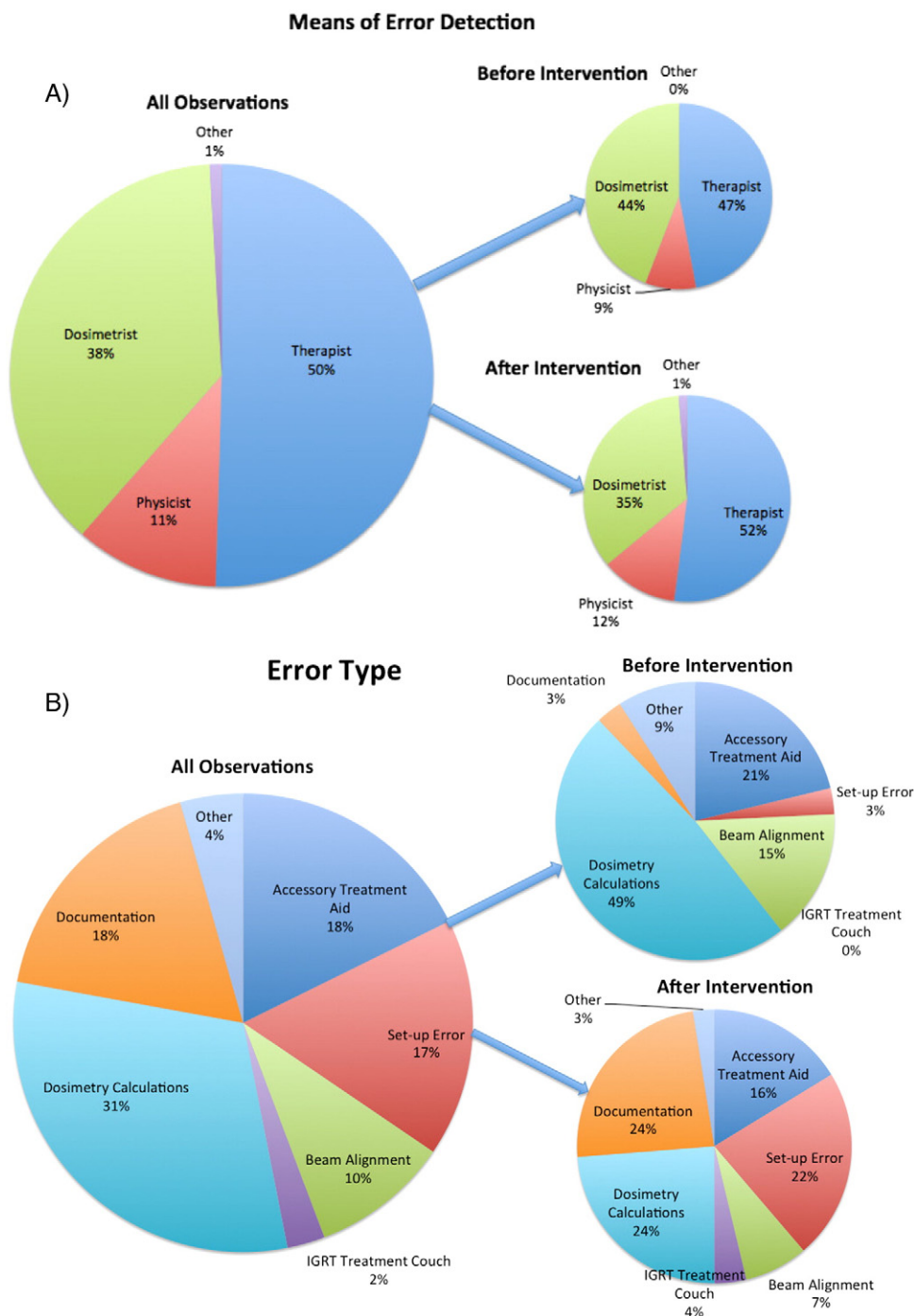


Figure 2 Pie charts illustrating patterns of error detection (A) and patterns of error type (B) before and after policy implementation. IGRT, image guided radiation therapy.

years before and after policy implementation. A statistically significant improvement remained.

When looking at the distribution of error reporting by staff, we show that the vast majority of error reporting was generated by therapists and dosimetrists. Before the initiation of the no-LSR policy, we observed that 44% of

errors were reported by dosimetrists compared with 35% of the errors following the policy. The reduction in error reporting from dosimetrists following the policy implementation is intuitive because dosimetrists typically bear the brunt of the burden from LSR. For the entire period, therapists accounted for half of all error reports; this is

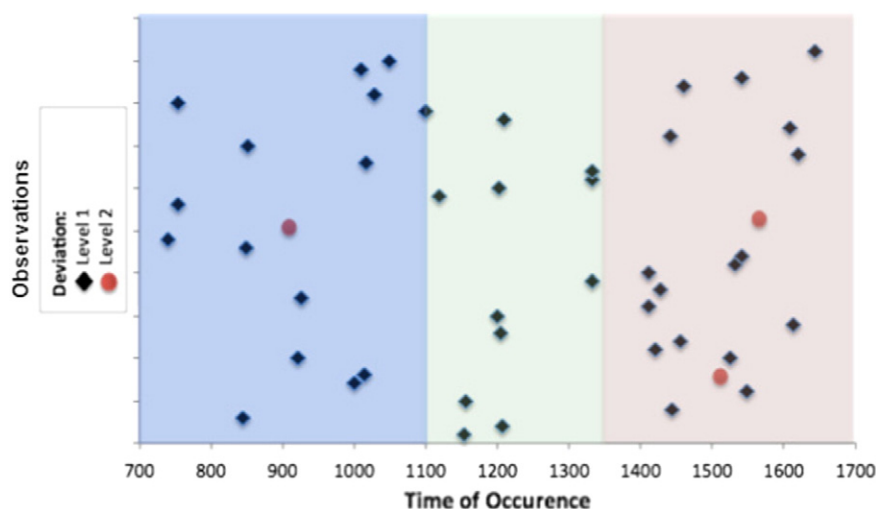


Figure 3 Plot showing primary treatment delivery error and time of day. Black corresponds to a Radiation Error Scoring System (RESS) level I error; red corresponds to a RESS level II error. There was no significant correlation between time of day and error rate ($P = .631$).

consistent with other studies that showed a relative level of comfort among therapists to report errors without fear of reprimand or negative repercussion.¹³

Although our analysis was focused on mandatory delivery error reporting, we did also track voluntary near-miss event reporting, which occurred infrequently. Through this process, we came to recognize a potential weakness of our error reporting system, which does not require near-miss reporting. Had near-miss reporting been incentivized or mandatory, one would expect significantly higher rates of near misses compared with errors reaching delivery. Of note, there were no nurse-reported errors and just one physician-reported error documented via this system. These discrepancies reinforce the need for a nonpunitive system in which all members of the treatment team can securely report both near-miss and treatment errors.

In this series, skin cancer and breast cancer treatments appeared most susceptible to treatment error. Although only 1.3% of patients were treated for skin cancer, this category accounted for 4.6% (5/109) of the treatment errors. Sixteen percent of patients were treated for breast cancer, but accounted for an inordinate 41% (45/109) of the treatment errors (Table 2). These results are not necessarily unexpected given the findings published by Walker et al. On multivariable analysis, their group found the number of prescription items (used as a surrogate for treatment complexity) to be a significant predictor of error.⁷ One would expect skin, breast, or postmastectomy prescriptions to potentially contain a high number of items—for example bolus, medial, and lateral tangent fields, supraclavicular field, posterior axillary boost field, electron field, or election boost—that is often prescribed to less than the 100% isodose line. In this series, incorrect use of bolus material accounted for 13 of the 45 errors affecting breast cancer patients.

Weekend simulation and treatments may be especially susceptible to error for several reasons, including variables in patient characteristics and treatment team. Patients requiring emergent weekend treatments typically have presented with some type of radiation emergency, including but not limited to, acute blood loss anemia as a result of hemoptysis or hematemesis, spinal cord compression, or superior vena cava syndrome. Patients presenting with these clinical symptoms may be particularly difficult to set up and often are uncomfortable in the typical supine treatment position. On-call dosimetrists and therapists may be more accustomed to planning or treating patients at a satellite facility or working as a member of a specific clinical team. In this series, we noted our error rates to be significantly higher for weekend simulation and treatments: 1.3% versus 0.09% per fraction ($P < .001$). Three errors were the result of dosimetric mistakes: 1 error in the transfer of plan data and 2 errors related to patient setup. Although the absolute number of weekend treatments in the series was low, the most common source of error was a dosimetric miscalculation, highlighting the need for enhanced quality assurance before plan delivery for emergent treatments.

Conclusions

To our knowledge, this is the first report describing the results of a specific policy designed to mitigate errors stemming from machine downtime, a dilemma that universally affects radiation therapy departments. Our delivery error rates compare favorably with published series from other academic institutions and were further improved following the implementation of a no large-scale

replanning policy. A higher error incidence with skin and breast cancer treatment delivery, as well as a significantly higher error rate for weekend simulation and treatment, were noted, warranting consideration of additional quality assurance measures in these areas.

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